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ITC-C508 / LAB EXERCISE WW-M2 / RESEARCH PROPOSAL

Evaluating a RAG-Based Chatbot for JRU CSE Program and Curriculum Inquiries

DecisionThree

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CHED AUTONOMOUS STATUS

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The Problem and Target Users

Target users

Graduating Grade 12 students considering JRU CSE programs.

Potential difficulty

Finding and understanding program details may require navigating lengthy documents.

Local needs check

An informal inquiry walkthrough will test whether students experience this difficulty.

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SDG 4: Quality Education

Target 4.3 concerns equal access to quality tertiary education. [1]

Clear program information can help prospective students understand their educational options before enrollment.

Research on college-major information provides context. JRU-specific needs still require validation. [2]

Measured contribution: accurate answers supported by official documents.

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Narrowing the Research Focus

Broad domain	Access to tertiary-education information
SDG-related issue	Difficulty locating and understanding program information
Target users	Graduating Grade 12 students considering JRU CSE
Narrow problem	Specific questions about programs and curriculum requirements
Chatbot focus	Inquiries answered using approved JRU documents

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Chatbot Focus and Example Inquiries

Program information

What undergraduate programs does JRU's CSE offer?

What is BS Information Technology about?

What does the Business Analytics specialization cover?

Curriculum requirements

What subjects are listed in first-year Computer Engineering?

How many units does this subject have?

What prerequisites are listed for this course?

Does Game Development include an internship?

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Scope and Limitations

Included

Program descriptions and specializations

Curriculum subjects and units

Documented course descriptions and prerequisites

Documented internship and project requirements

Outside the initial scope

Tuition quotations and payment concerns

Application processing or status tracking

Live schedules and faculty availability

Personalized recommendations

Later curriculum editions without sources

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Primary Research Question

How accurately and with what level of documentary support can a RAG-based chatbot answer graduating Grade 12 students' inquiries about JRU CSE programs and curriculum requirements?

Correctness

Factual accuracy against verified reference answers.

Groundedness

Official documents support the answer's factual claims.

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Proposed Knowledge-Base Sources

Primary source [3]

JRU College General Information, School Year 2025–2026.

Supporting sources [4–9]

Official CSE listing and relevant program profiles, subject to version checks.

Source audit

Record program, specialization, edition, page numbers, applicability, and conflicts.

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Supporting Question and Test Design

How do answer correctness and groundedness differ between the RAG-based chatbot and the same language model without retrieval augmentation?

With retrieval

Model receives the question and passages retrieved from approved documents.

Without retrieval

The same model receives the question without source passages or document access.

Controls: same model configuration, questions, generation settings, and answer limit.

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Evaluation Question Set

30

Fully answerable

Five questions in each of six inquiry categories.

5

Partly answerable

Documents support only part of the requested answer.

5

Unsupported in scope

Requested information is absent from approved sources.

Approximately 40 questions and 80 responses across the two conditions.

Faculty-reviewed answer key and two independent scorers. Separate out-of-scope checks.

Evaluation Measures

Primary measures

Correctness
Share of factual claims matching the answer key.

Groundedness
Share of claims supported by approved documents.

Supporting measures

Completeness of required reference facts.

Citation support in RAG responses.

Appropriate handling of missing information and unnecessary refusals.

Report paired differences, category results, scorer agreement, and errors.
A refusal receives no automatic full credit for correctness or groundedness. [11, 12]

Proposed Model and Manageability

Qwen3-8B in non-thinking mode [13]

Preliminary choice with an Apache 2.0 license. A hardware pilot will confirm feasibility. Both conditions will use the same checkpoint and configuration.

Work sequence

Needs check and source audit. Hardware and retrieval pilot. Prototype and answer-key review. Configuration freeze. Evaluation and reporting.

Proposal stage: implementation and performance results remain future work.

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